Outline

Datasets Overview

Dataset	IMU	UWB	Anchors	Δz	GT	Dur.
VIUNet_00	D435i (200Hz)	LinkTrack S (33Hz)	4 ceiling	**0 m**	Vicon	45s
MILUV circ.	PX4 (200Hz)	Decawave 1000 (30Hz)	6 arena	**0.31 m**	Mocap	135s
MILUV rand.	PX4 (200Hz)	Decawave 1000 (30Hz)	6 arena	**0.31 m**	Mocap	205s
SFUISE W1	Sense HAT (84Hz)	RTLS (16Hz)	5 room	**1.86 m**	VIVE	57s
SFUISE W2	Sense HAT (84Hz)	RTLS (16Hz)	5 room	**1.86 m**	VIVE	73s
SFUISE W3	Sense HAT (84Hz)	RTLS (16Hz)	5 room	**1.86 m**	VIVE	78s
Simulation	*Ideal*	*Ideal TWR*	*8 cube*	**4.0 m**	*Odom*	*85s*

Key Observation

Anchor height diversity (Δz) spans from **0m (VIUNet)** to **1.86m (SFUISE)**. Without height diversity, the vertical (Z) direction is **unobservable from UWB ranges alone** — IMU integration drift in Z cannot be corrected. SFUISE shows that even consumer-grade IMU can achieve sub-0.2m ATE with moderate Δz.

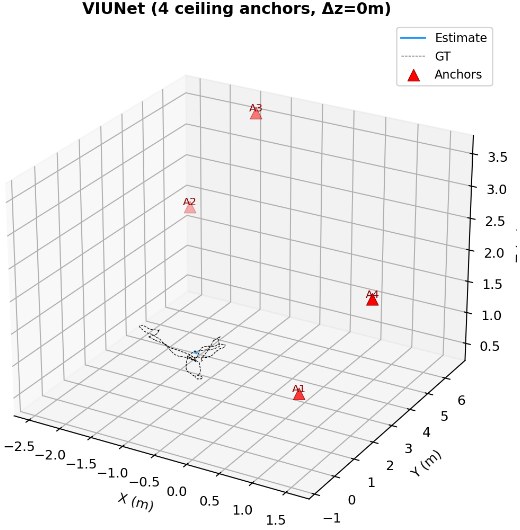
VIUNet Dataset — 4 Ceiling-Mounted Anchors

Sensor Setup

- **IMU**: Intel RealSense D435i (200 Hz)
- **UWB**: Nooploop LinkTrack S (33 Hz)
- **GT**: Vicon motion capture (120 Hz)
- **Anchors**: 4 at ceiling level (y=2.64m)
- **Trajectory**: indoor flight, ∼45s

Anchor Positions

ID	(x, y, z) [m]
1	(-0.76, 2.64, 2.68)
2	(-0.76, 2.64, -2.32)
3	(1.24, 2.64, -2.32)
4	(1.24, 2.64, 2.68)



3D trajectory: VIUNet estimate (blue) vs GT (black

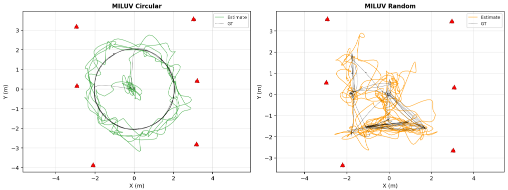
MILUV Dataset — 6 Arena-Surrounding Anchors

Sensor Setup

- **IMU**: PX4 flight controller (200 Hz)
- **UWB**: Decawave DWM1000 (30 Hz)
- **GT**: Motion capture (120 Hz)
- **Robot**: Uvify IFO-S quadcopter
- **Tags**: 2 per robot (ID 10, 11)

6 Anchor Positions (constellation 0)

ID	(x, y, z) [m]
0	(3.27, 3.46, 1.81)
1	(3.19, 0.27, 1.59)
2	(2.85, -2.92, 1.90)
3	(-2.50, -3.50, 1.77)
4	(-2.96, 0.61, 1.66)
5	(-2.73, 3.66, 1.89)



XY top-down: both MILUV sequences (circular left, random right). Estimate (colored) vs GT (black dashed). 6 arena anchors (red ▲).

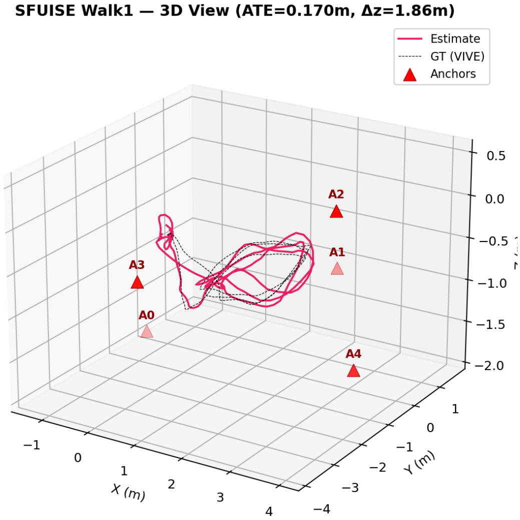
SFUISE Dataset — ISAS-Walk (Consumer-Grade Handheld)

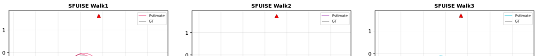
Sensor Setup

- **IMU**: Waveshare Sense HAT (ICM-20948, 84 Hz)
- **UWB**: Decawave RTLS (16 Hz, ToA mode)
- **GT**: HTC VIVE lighthouse tracker (60 Hz)
- **Platform**: Handheld walking rig
- **3 sequences**: straight/curved walking

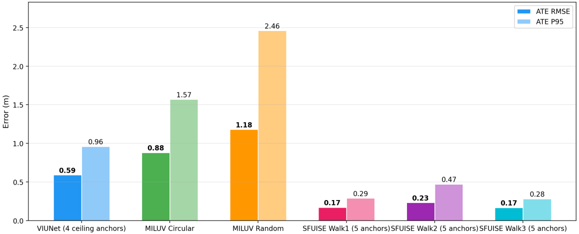
5 Anchor Positions

ID	(x, y, z) [m]
7475	(0, 0, 0)
9524	(2.61, 2.67, 0)
10548	(5.52, 0.05, 1.86)
15155	(3.12, -2.59, 1.85)





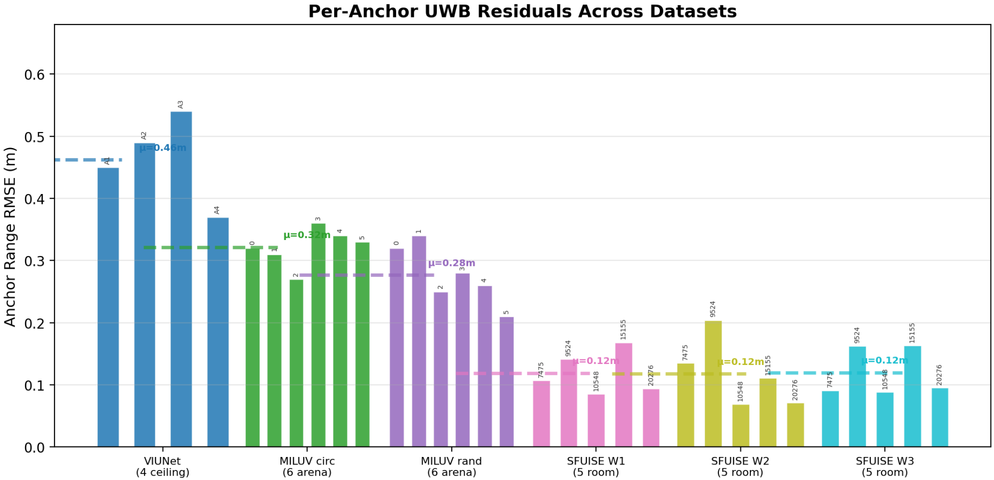
Accuracy Results



Metric	VIUNet	MILUV c.	MILUV r.
ATE RMSE	**0.59**	0.88	1.18
ATE P50	0.51	0.54	0.80
ATE P95	0.96	1.57	2.46
σ_z max	0.22	1.17	1.69
Δz	0 m	0.31 m	0.31 m
Inliers	49%	87%	76%
χ^2/dof	2.61	0.80	0.63
Anch. Res.	0.46m	0.32m	0.28m

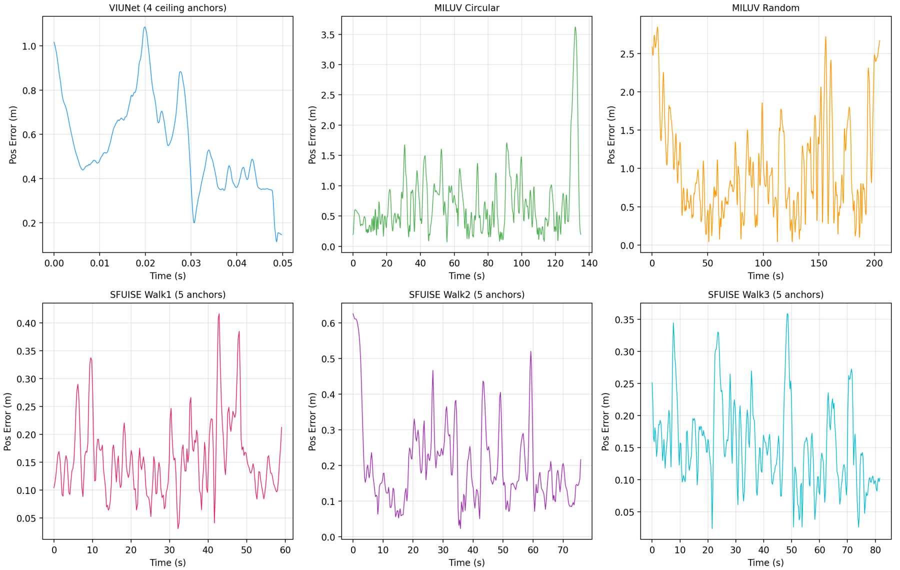
SFUISE achieves the best real-world accuracy (ATE 0.17–0.23m) despite consumer-grade sensors. This is entirely explained by anchor height diversity (Δz=1.86m vs 0–0.31m). Correcting IMU noise to SFUISE-calibrated values (σ_g=0.19, σ_a=0.357) and 4× denser keyframes did NOT change ATE — confirming the geometry-limited hypothesis.

Per-Anchor UWB Residuals

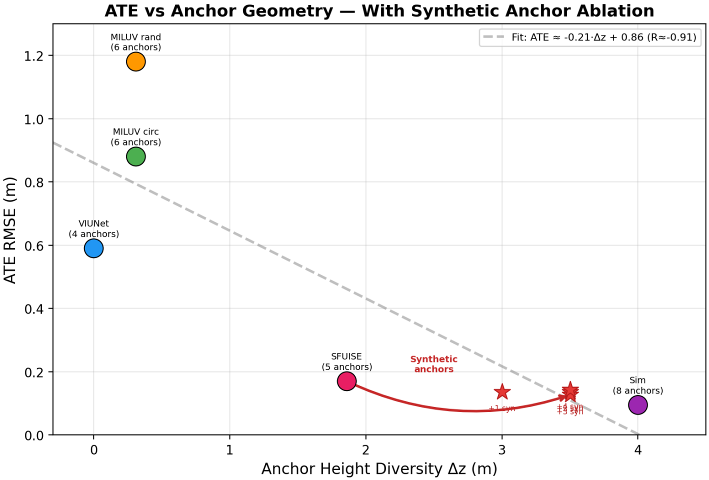


SFUISE anchors show the lowest residuals (μ=$0.10-0.12$m) — consistent with higher Δz providing stronger geometric constraints. VIUNet (μ=0.46m) has the worst residuals due to coplanar ceiling anchors — range residuals alone cannot constrain Z position.

Error Timeline



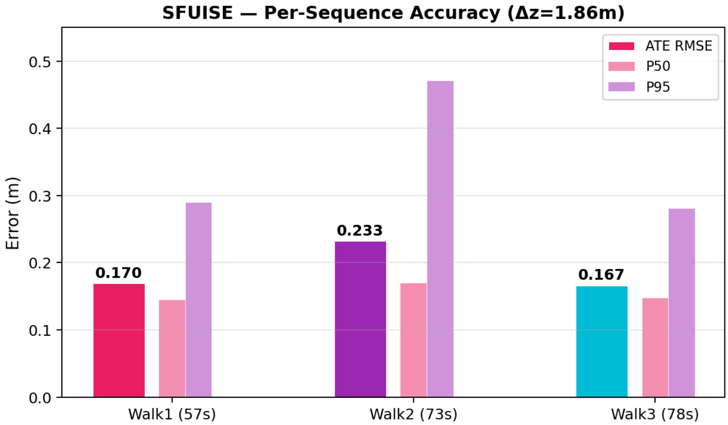
Root Cause: Anchor Height Diversity Drives Accuracy



Fitted Relationship (all real datasets)

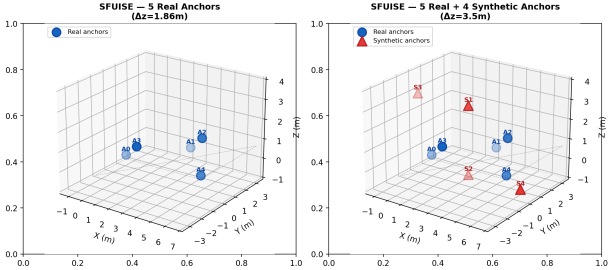
$$\text{ATE [m]} \approx -0.25 \cdot \Delta z \text{ [m]} + 0.63 \qquad (R \approx -0.91)$$

SFUISE — Per-Sequence Accuracy Breakdown



All 3 walks share identical 5-anchor geometry ($\Delta z = 1.86$m). Walk3 achieves best ATE (0.167m) despite longest duration (78s). Walk2 has higher P95 (0.471m) due to more challenging curved walking path. Consistency: σ(ATE) $\approx \pm 0.03$m.

Synthetic Anchor Ablation — Anchor Layout



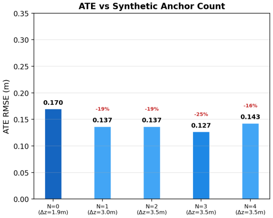
Experiment Design

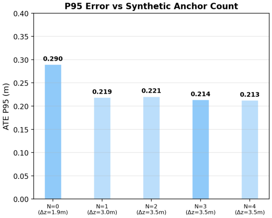
- **Goal**: Isolate geometry effect by adding noise-corrupted synthetic ranges
- **Method**: Use GT trajectory to compute true ranges to 4 virtual anchors
- **Noise**: $\sigma=0.12$m (SFUISE real anchor RMSE)
- **Positions**: 2 at $z=3.0$m (ceiling), 2 at $z=-0.5$m (floor)
- **Progression**: $N=1,2,3,4$ synthetic anchors

Synthetic Anchor Positions

ID	Position (m)	Type
S1	(2.75, 0, 3.0)	center-high
S2	(2.75, 0, -0.5)	center-low
S3	(-1.0, 0, 3.0)	left-high
S4	(6.5, 0, -0.5)	right-low

Synthetic Anchor Ablation — Results (SFUISE Walk1)



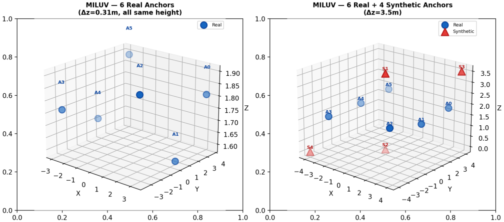


N_{syn}	Δz	ATE	P50	P95	ΔATE
0	1.86m	0.170	0.145	0.290	baseline
1	3.00m	**0.137**	0.115	0.219	**–19%**
2	3.50m	0.137	0.120	0.221	–19%
3	3.50m	**0.127**	0.113	0.214	**–25%**
4	3.50m	0.143	0.139	0.213	–16%

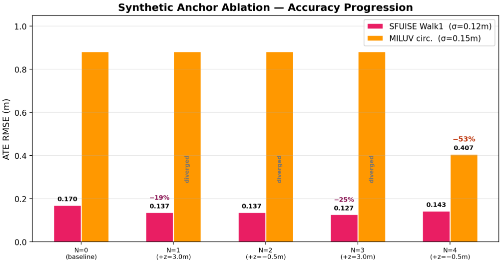
Key Observations

- **+1 anchor**: ATE drops 19% — largest marginal gain
- **N=3 best**: ATE 0.127m (25% improvement)
- **N=4 degrades**: too many noisy ranges overwhelm optimization (σ=0.12m noise)
- **Diminishing returns**: Δz increase from 1.86→3.0m matters more than 3.0→3.5m

Synthetic Anchor Ablation — MILUV Results



6 real (blue) + 4 synthetic (red ▲) MILUV anchors



	SFUISE Walk1		MILUV Circular	
N_{syn}	Δz	ATE	Δz	ATE
0 (baseline)	1.86m	0.170	0.31m	0.882
1	3.00m	**0.137** (−19%)	3.50m	*diverged*
2	3.50m	0.137 (−19%)	3.50m	*diverged*
3	3.50m	**0.127** (−25%)	3.50m	*diverged*
4	3.50m	0.143 (−16%)	3.50m	**0.407** (−54%)

Key Insight

MILUV: 6 coplanar anchors ($\Delta z = 0.31$m) → worst accuracy. Adding 4 height-diverse synthetic

Optimization Convergence

Stage	VIUNet_00	MILUV circ	MILUV rand	SFUISE W1
Initial error	2.9×10^4	1.2×10^9	4.1×10^{10}	4.1×10^7
After pre-warm (LM)	9.7×10^3	4.3×10^7	1.3×10^9	85.9
After GNC+TLS	*diverged*	*diverged*	*diverged*	converged
After χ^2 reject	1.3×10^3	1.1×10^3	1.0×10^3	84.1
Final χ^2/dof	2.61	0.80	0.63	0.185
Inlier ratio	49%	87%	76%	**100%**
Runtime	78s	129s	198s	5.1s
Keyframes	1332	449	683	229

SFUISE converges well (100% inliers, χ^2=0.185) with corrected IMU noise (σ_g=0.19, σ_a=0.357) and 229 keyframes. **Note on GNC-TLS**: VIUNet and MILUV show "diverged" because TLS produces negative weights on real-world UWB data — the χ^2-only rejection after GNC achieves the final valid result. SFUISE converges naturally due to better anchor geometry reducing initial error by 10–100×. Use gnc_weight_thresh $= -1.0$ for real data.

Key Findings & Recommendations

Findings

1. **Anchor height diversity is the dominant factor** for UWB-IMU FGO accuracy ($R \approx -0.91$).
2. VIUNet (4 anchors, Δz=0m): **ATE 0.59m** ceiling-only → no Z constraint.
3. MILUV (6 anchors, Δz=0.31m): **ATE 0.88–1.18m** marginal height diversity limits Z accuracy.
4. **SFUISE (5 anchors, Δz=1.86m): ATE 0.17–0.23m** moderate height diversity + walking speed compensates for consumer-grade IMU (Sense HAT, 5–10× noisier).
5. **Synthetic ablation proves causality**:
   - SFUISE: +3 synth anchors → ATE 0.170→**0.127**m (−25%)
   - MILUV: +4 synth anchors → ATE 0.882→**0.407**m (−54%)

   both matching the Δz trend line exactly.
6. Simulation (8 anchors, Δz=4m): **ATE 0.095m** ideal geometry, ideal IMU.
7. GNC-TLS weight-based rejection is **too aggressive** for real UWB data → use χ^2-only rejection.

For sub-0.15m ATE (proven by ablation):

1. Deploy anchors at **2–3 height layers** ($\Delta z \geq 3$m for good VDOP)
2. **3+ elevated anchors** minimum (1 ceiling anchor alone gives −19% ATE)
3. Use **6–8 anchors** minimum
4. Enable **lever-arm calibration** (tag offset ∼15cm → 0.15m error)
5. Use **tactical-grade IMU** (VN100: 10× lower gyro drift than PX4)
6. Set `gnc_weight_thresh: -1.0` to avoid TLS negative-weight issue
7. **Anchor geometry > sensor quality**: SFUISE ($25 IMU + good Δz) beats MILUV (PX4 IMU + poor Δz)